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ROBOTIC END EFFECTOR INCLUDING A NODE

Abstract

A robotic end effector includes a node having at least one wall. Each wall has a corresponding interior side and a corresponding exterior side. The interior side of the wall defines a closed profile disposed about an axis such that the interior side of the wall defines a cavity and the wall defines an end aperture open to the cavity through an axial end of the node. The end aperture being configured to receive a frame member. The wall includes a port having an outlet open through the interior side of the wall and an inlet open through the exterior side of the wall. The interior side of the wall defines a plurality of recessed channels that intersect the outlet of the port.

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Background/Summary

FIELD

[0001] The present disclosure relates to a robotic end effector including a node.

BACKGROUND

[0002] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0003] Industrial robots have been used for a variety of manufacturing operations, including by way of example, welding, placement of parts for subsequent fabrication or assembly operations, and moving parts from one location to another such as retrieving parts from a storage location and moving them to an assembly station. These industrial robots include end effectors, which are essentially the hands of the robot. These end effectors come in a variety of configurations depending on the particular manufacturing operation. However, end effectors often are large and heavy, and lack mechanical repeatability.

[0004] These issues related to robotic end effectors, among other issues related to robotic end effectors, are addressed by the present disclosure.

SUMMARY

[0005] This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

[0006] In one form, the present disclosure provides a robotic end effector that includes a first node. The first node includes at least one wall. The wall has a corresponding interior side and a corresponding exterior side. The interior side of the wall defines a closed profile disposed about an axis such that the interior side of the wall defines a cavity and the wall defines a first end aperture open to the cavity through an axial end of the first node. The first end aperture is configured to receive a first frame member. The wall includes a first port having a first outlet open through the interior side of the first wall and a first inlet open through the exterior side of the first wall. The interior side of the first wall defines a plurality of first recessed channels that intersect the first outlet of the first port.

[0007] In variations of the robotic end effector of the above paragraph, which can be implemented individually or in any combination: the first node is made of plastic; the interior side of the first node defines an interior surface, the first recessed channels are formed in the interior surface; the first node includes a locating feature that is adjacent the first port and configured to facilitate positioning of the first node relative to the frame, the locating feature includes a second outlet open through the interior side of the first node and a second inlet end open through to the exterior side of the first node; the first recessed channels are spaced apart from the second outlet of the locating feature and extend around the second outlet of the locating feature; a second port that includes a second outlet open through the interior side of the first node and a second inlet open through the exterior side of the first node; second recessed channels formed in the interior side of the first node and intersecting the second outlet of the second port; the second recessed channels have a pattern that is different from a pattern of the first recessed channels; the first node includes a mounting feature that is adjacent to the first port and configured to mount a structure to the first node; the mounting feature includes a plurality of threaded apertures; the robotic end effector further includes the frame, w a bonding agent flows through the first port to the first recessed channels to attach the first node to the frame; and a second node that is spaced apart from the first node and attached to the frame.

[0008] In another form, the present disclosure provides a robotic end effector that includes a frame and a first node. The frame includes a first member and a second member that extends in a direction that is different than the first member. The first node includes a first interior side, a second interior side, and an exterior side. The first interior side faces the first member and defines a first interior cavity that receives the first member. The second interior side faces the second member and defines a second interior cavity that receives the second member. The exterior side faces away from the frame. The first node further includes a first port, first recessed channels, a second port and second recessed channels. The first port includes a first outlet that opens through to the first interior side of the first node and a first inlet that opens through to the exterior side of the first node. The first recessed channels formed in the first interior side of the first node are proximate the first member and intersects the first outlet of the first port. The second port includes a second outlet that opens through to the second interior side of the first node and a second inlet that opens through to the exterior side of the first node. The second recessed channels are formed in the second interior side of the first node proximate the second member and intersecting the second outlet of the second port.

[0009] In variations of the robotic end effector of the above paragraph, which can be implemented individually or in any combination: the second recessed channels form a pattern that is different from a pattern of the first recessed channels; the robotic end effector includes a second node that is spaced apart from the first node and attached to the frame; the first node includes a mounting feature that is adjacent to the first port and configured to mount a structure to the first node; the first node further includes a third port and an internal port; the third port includes a third outlet and a third inlet, the third inlet opens through to the exterior side of the first node; the internal port is located within the first node and extends from the first interior side of the first node to the second interior side of the first node, the internal port in fluid communication with the third outlet of the third port; the internal port includes a diameter that is greater than a diameter of the third port; the first node includes a locating feature that is adjacent to one of the first and second ports and that facilitates positioning of the first node relative to the frame, the locating feature includes a third outlet that opens through to one of the first and second interior sides of the first node and a third inlet that opens through to the exterior side of the first node; and one of the first and second recessed channels are spaced apart from the third outlet of the locating feature and extend around the third outlet of the locating feature.

[0010] In yet another form, the present disclosure provides a method of assembling a robotic end effector. The method includes inserting a first frame member into a first interior cavity of a node such that first recessed channels defined by a first interior surface of the first interior cavity are proximate the first frame member; inserting a second frame member into a second interior cavity of the node such that second recessed channels defined by a second interior surface of the second interior cavity are proximate the second frame member; injecting a first bonding agent into a first port of the node such that it flows along the first recessed channels in the first interior surface; and injecting a second bonding agent into a second port of the node such that it flows along the second recessed channels in the second interior surface.

[0011] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

[0012] In order that the disclosure may be well understood, there will now be described various forms thereof, given by way of example, reference being made to the accompanying drawings, in which:

[0013] FIG. **1** is a perspective view of a robot including a robotic end effector secured to a workpiece in accordance with the principles of the present disclosure;

[0014] FIG. **2** is a perspective view of the robotic end effector of FIG. **1** secured to the workpiece;

[0015] FIG. **3** is a top view of the robotic end effector of FIG. **1**;

[0016] FIG. **4** is a perspective view of the robotic end effector of FIG. **1**;

[0017] FIGS. **5**A-**5**H are perspective views of various nodes of the robotic end effector of FIG. **1**;
[0018] FIG. **6** is a cross-sectional view of the robotic end effector of FIG. **1** taken along line **6-6** of FIG. **4**;
[0019] FIG. **7** is a cross-sectional view of the robotic end effector of FIG. **1** taken along line **7-7** of FIG. **4**;
[0020] FIG. **8** is a cross-sectional view of one robotic end effector of FIG. **1**;
[0021] FIG. **9** is a perspective view of one robotic end effector of FIG. **1**;
[0022] FIG. **10** is a cross-sectional view of one robotic end effector of FIG. **1**; and
[0023] FIG. **11** is a flowchart depicting a method for assembly the robotic end effector of FIG. **1**.
[0024] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

DETAILED DESCRIPTION

[0025] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

[0026] With reference to FIG. **1**, a robot **10** for securing and moving a workpiece or part **12** is provided. In the example illustrated, the workpiece **12** is a stamped plate. However, the workpiece **12** may be other suitable objects such as a component built using another manufacturing process (e.g., forming, joining, extrusion, casting), or an additive manufacturing process (i.e., a 3-D printing). In the example illustrated, the robot **10** is positioned on and secured to a fixed surface (e.g., a ground surface of a manufacturing facility) near a workstation (not shown). However, in some configurations, the robot **10** may be positioned on a movable platform (i.e., the platform moves to the workstation where the robot **10** grasps and moves the workpiece **12** from a rack, for example, to the workstation). In such configuration, the platform may be moved automatically or manually.

[0027] The robot **10** includes a robot arm **14** and a robotic end effector **16**. The robot arm **14** includes a plurality of segments connected to each other at joints, thereby allowing the robot arm **14** to have multiple degrees of freedom. The robot arm **14** is also secured to the fixed surface at a first end **14***a*. In some variations, the robot arm **14** includes an optional adapter (not shown) that is adapted to be secured to the fixed surface.

[0028] With reference to FIGS. **2-4**, the robotic end effector **16** is used to secure or otherwise attach the workpiece **12** (FIG. **2**) to the robot arm **14** and includes a frame **18** and a plurality of nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f*. In one form, the frame **18** may be manufactured using traditional manufacturing processes such as casting, joining or forming, for example. In another form, the frame **18** may be manufactured using an additive manufacturing process (i.e., 3-D printing). The frame **18** may be made of a carbon fiber material, for example, and may include a pair of rails or frame members **22***a*, **22***b* and a plurality of cross members **24***a*, **24***b*, **24***c*. The pair of frame members **22***a*, **22***b* are spaced apart from each other and extend along a longitudinal direction X of the robotic end effector **16**. In the example illustrated, each frame member **22***a*, **22***b* has a rectangular shape and includes a plurality of sides that cooperate with each to define a cavity (not specifically shown). In some forms, the frame members **22***a*, **22***b* may have a circular shape, a square shape, or any other suitable shape.

[0029] Each cross member **24***a*, **24***b*, **24***c* has a rectangular or square shape and includes a plurality of sides that cooperate with each to define a cavity. In some forms, the cross members **24***a*, **24***b*, **24***c* may have a circular shape, a pentagonal shape, or any other suitable shape. Each cross member **24***a* extends in a transverse direction Y of the robotic end effector **16** between the pair of frame members **22***a*, **22***b* and is connected to the pair of frame members **22***a*, **22***b* by the nodes **20***e*, which will be described in more detail below.

[0030] Each cross member **24***b* extends in the transverse direction Y past the pair of frame members **22***a*, **22***b* and is connected to the pair of frame members **22***a*, **22***b* by the nodes **20***b*, which will be described in more detail below. Each cross member **24***c* extends in a transverse direction Y past the pair of frame members **22***a*, **22***b* and is connected to the pair of frame members **22***a*, **22***b* by the nodes **20***c*, which will be described in more detail below. In the example illustrated, the cross member **24***c* is located below the pair of frame members **22***a*, **22***b* and below the cross members **24***a*, **24***b*.

[0031] The nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f* connect parts of the frame **18** to each other and/or connects the frame **18** to a separate structure (e.g., a securing or attachment structure). For example, each node **20***e* connects a respective frame member **22***a*, **22***b* to a respective cross member **24***a* and/or connects the frame **18** to the robot arm **14** via node **20***f*. In another example, each node **20***a* connects the frame **18** to an attachment structure **60** that attaches or otherwise secures the workpiece **12**. In yet another example, each node **20***c* connects a respective frame member **22***a*, **22***b* to the cross member **24***c*. Each node **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f* is in the form of a single unitized, monolithic body that can be manufactured by an additive manufacturing process and secured to the frame **18**. The manufacturing process can include laser sintering, for example, that can generally include a laser (not shown), a device (not shown) for applying subsequent layers of powdered sintering material (e.g., polyamide powder), and a controller (not shown) that controls operation of the laser and the amount and timing of the deposition of the polyamide powder. It should be understood that other 3D printing or additive manufacturing methods may be employed to achieve the unitized, monolithic body, along with a variety of different materials, while remaining within the scope of the present disclosure. In some forms, the nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f* may be manufactured directly onto the frame **18** using at least one additive manufacturing technique or process, such as, for example, directed energy deposition (DED), wire fed arc weld, powder-laser directed energy deposition, or cold spray.

[0032] As shown in FIGS. **3** and **4**, the nodes **20***a* are secured to various parts of the frame **18**. That is, one or more nodes **20***a* may be secured to a respective frame member **22***a*, **22***b* and/or one or more nodes **20***a* may be attached to a respective cross member **24***a*, **24***b*, **24***c*. With reference to FIGS. **5**A, **6**, and **9**, each node **20***a* includes a plurality of walls **34***a* **34***b*, **34***c*, **34***d* that cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is rectangular. In some forms, the node **20***a* includes one or more walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each wall **34***a*, **34***b*, **34***c*, **34***d* has an interior side **36** and an exterior side **38**. The closed profile is disposed about an axis A (FIG. **5**A) such that the interior sides **36** of the walls **34***a*, **34***b*, **34***c*, **34***d* cooperate with each other to define a cavity **40** and at least one end aperture **42** open to the cavity **40** through respective axial end of the node **20***a*. In this way, the end apertures **42** may receive a portion of the frame **18**.

[0033] Each node **20***a* further includes a plurality of ports **46**, a plurality of locating features **50**, and at least one mounting feature **52**. A bonding agent (e.g., adhesive such as epoxy) may be injected into the ports **46** to bond the node **20***a* to the frame **18** as described in more detail below. In the example illustrated, each wall **34***a*, **34***b*, **34***c*, **34***d* includes at least one port **46** having an outlet open through the interior side **36** of the wall **34***a*, **34***b*, **34***c*, **34***d* and an inlet open through the exterior side **38** of the wall **34***a*, **34***b*, **34***c*, **34***d*. In some forms, one wall **34***a*, **34***b*, **34***c*, **34***d* or less than all walls **34***a*, **34***b*, **34***c*, **34***d* may include the port **46** formed therein. An interior surface **36***a* of the interior side **36** of each wall **34***a*, **34***b*, **34***c*, **34***d* defines a plurality of recessed channels **48** that intersect the outlet of the port **46**. In one configuration, the recessed channels **48** in the interior side **36** of one wall **34***a*, **34***b*, **34***c*, **34***d* may form a pattern that may be different from a pattern formed in the recessed channels **48** in the interior side **36** of another wall **34***a*, **34***b*, **34***c*, **34***d*. The recessed channels **48** in the interior side **36** of each wall **34***a*, **34***b*, **34***c*, **34***d* may include arcuate channels, linear channels, inclined channels, or a combination of arcuate, linear, and/or inclined channels.

[0034] The locating features **50** may facilitate positioning and holding of the node **20***a* onto the frame **18** prior to bonding the node **20***a* onto the frame **18** using the bonding agent. The locating features **50** are apertures formed in the node **20***a*. Each locating feature **50** includes an outlet open through the interior side **36** of the respective wall **34***a*, **34***b* and an inlet open through the exterior side **38** of the respective wall **34***a*, **34***b*. In the example illustrated, one locating feature **50** is formed in the wall **34***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one locating feature is formed in the wall **34***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. In this way, a pin (not shown) may extend through each locating feature **50** and the corresponding aperture to temporarily hold the location of the node **20***a* on the frame **18** until the bonding agent bonds the node **20***a* to the frame **18**. In some configurations, locating features **50** may be formed in the walls **34***c*, **34***d*, instead of, or in addition to, the locating features **50** formed in the walls **34***a*, **34***b.*

[0035] In the example illustrated, the locating feature **50** on the wall **34***a* is adjacent to a respective port **46** on the wall **34***a* and the locating feature **50** on the wall **34***b* is adjacent a respective port **46** on the wall **34***b*. It should be understood that the recessed channels **48** in the interior side **36** of the wall **34***a*, **34***b* may be spaced apart from the outlet of the locating feature **50** and/or extend around the outlet of the locating feature **50**. In this way, the bonding agent injected into the port **46** is inhibited from flowing out of the locating feature **50** onto the exterior side **38** of the walls **34***a*, **34***b*. In some forms, the locating feature **50** on the wall **34***a* and the respective port **46** on the wall **34***a* may be located at opposite ends of the wall **34***a*, and locating feature **50** on the wall **34***b* and the respective port **46** on the wall **34***b* may be located at opposite ends of the wall **34***b.*

[0036] The mounting feature **52** is configured to secure a structure **60** (FIG. **2**) onto the first node **20***a*. In the example illustrated, the structure **60** is configured to attach or otherwise secure the workpiece **12** to the robotic end effector **16**. In one example, the structure **60** is mounted to the mounting feature **52** and includes one or more magnets (not specifically shown). The magnets may be energized such that the workpiece **12** is attached to the robotic end effector **16** and may be de-energized such that the workpiece **12** is detached from the robotic end effector **16**. In some forms, the structure **60** may include hooks, gripping arms, or another suitable structure to grasp, hold or otherwise secure the workpiece **12** onto the robotic end effector **16** such that the workpiece **12** can be moved from one location to another, for example.

[0037] In the example illustrated, the mounting feature **52** is formed on the exterior side **38** of the wall **34***d* of the node **20***a*. In some forms, the mounting feature **52** may be formed on the exterior side **38** of another wall **34***a*, **34***b*, **34***c*, instead of, or in addition to, being formed on the exterior side **38** of the wall **34***d*. The mounting feature **52** may include a surface **54** that is raised from an exterior surface **38***c* or otherwise located a predetermined distance from the exterior surface **38***c*. The surface **54** includes a plurality of apertures **56** formed therein. In the example illustrated, an adapter **58** (FIG. **2**) is mounted to the mounting feature **52** and the structure **60** is secured to the adapter **58**. In some forms, the structure **60** may be secured directly to the mounting feature **52**.

[0038] In some configurations, as shown in FIG. **3**, the robotic end effector **16** may include nodes **51***a* disposed on the cross members **24***a* between two nodes **20***e*. Each node **51***a* may include two mounting features **52***a*. In some configurations, the robotic end effector **16** may also include nodes **51***b* disposed on a respective frame member **22***a*, **22***b* that do not include mounting features. In such configuration, a support structure **55** may extend in the transverse direction Y and may be attached to the nodes **51***b* such that the robotic end effector **16** may be positioned or disposed on a stand or rack (not shown) via the support structure **55**.

[0039] With reference to FIGS. **4**, 5B, and **10**, the nodes **20***b* attach a respective frame member **22***a*, **22***b* to the cross member **24***b*. Each node **20***b* includes a plurality of first walls **62***a* **62***b*, **62***c*, **62***d* and a plurality of second walls **64***a*, **64***b*, **64***c*, **64***d*. The plurality of first walls **62***a*, **62***b*, **62***c*, **62***d* cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is rectangular. In some forms, each node **20***b* includes one or more first walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each first wall **62***a*, **62***b*, **62***c*, **62***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**1** such that the interior sides of the walls **62***a*, **62***b*, **62***c*, **62***d* cooperate with each other to define a cavity **70** and end apertures **72** (only one shown in FIG. **5**B) open to the cavity **70** through an axial end of the node **20***b*. In this way, the end aperture **72** may receive the cross member **24***b* of the frame **18**.

[0040] The plurality of second walls **64***a*, **64***b*, **64***c*, **64***d* cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the closed profile formed by the second walls **64***a*, **64***b*, **64***c*, **64***d* extend in a direction perpendicular from the closed profile formed by the first walls **62***a*, **62***b*, **62***c*, **62***d*. In some forms, the closed profile formed by the second walls **64***a*, **64***b*, **64***c*, **64***d* may extend in an oblique direction from the closed profile formed by the first walls **62***a*, **62***b*, **62***c*, **62***d*. In some forms, each node **20***b* may include one or more second walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each second wall **64***a*, **64***b*, **64***c*, **64***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**2** such that the interior sides of the second walls **64***a*, **64***b*, **64***c*, **62***d* cooperate with each other to define a cavity (not specifically shown) and at least one end aperture (not specifically shown) open to the cavity through a respective axial end of the node **20***b*. In this way, the end aperture may receive the respective frame member **22***a*, **22***b* of the frame **18**.

[0041] Each node **20***b* further includes a plurality of first ports **86***a*, a plurality of second ports **86***b*, first recessed channels (not shown), second recessed channels (not shown), a plurality of first locating features **88***a*, and a plurality of second locating features **88***b*. A bonding agent (e.g., adhesive such as epoxy) may be injected into the first ports **86***a* to bond the node **20***b* to the cross member **24***b* and a bonding agent may be injected into the second ports **86***b* to bond to the node **20***b* to the frame members **22***a*, **22***b*. In the example illustrated, each first wall **62***a*, **62***b*, **62***c*, **62***d* includes at least one first port **86***a* and each second wall **64***a*, **64***b*, **64***c*, **64***d* includes at least one second port **86***b*. The structure and function of the first and second ports **86***a*, **86***b* may be similar or identical to the ports **46** described above, and therefore, will not be described again in detail.

[0042] The first recessed channels (not shown) are formed in an interior side (not shown) of each first wall **62***a*, **62***b*, **62***c*, **62***d* and intersect an outlet (not shown) of the first port **86***a*. Similarly, the second recessed channels (not shown) are formed in an interior side (not shown) of each second wall **64***a*, **64***b*, **64***c*, **64***d* and intersect an outlet (not shown) of the second port **86***b*. The structure and function of the first and second recessed channels may be similar or identical to that of the recessed channels **48** described above, and therefore, will not be described again in detail.

[0043] In the example illustrated, one first locating feature **88***a* is formed in the first wall **62***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one first locating feature **88***a* is formed in the first wall **62***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. Similarly, one second locating feature **88***b* is formed in the second wall **64***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one second locating feature **88***b* is formed in the second wall **64***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. The structure and functions of the first and second locating features **88***a*, **88***b* may be similar or identical to the locating features **50** described above, and therefore, will not be described again in detail. Connecting members **89** may connect one or more of the first walls **62***a*, **62***b*, **62***c*, **62***d* to one or more of the second walls **64***a*, **64***b*, **64***c*, **64***d*.

[0044] With reference to FIGS. **4**, **50**, and **8**, the nodes **20***c* attach a respective frame member **22***a*, **22***b* to the cross member **24***c*. Each node **20***c* includes a plurality of first walls **92***a* **92***b*, **92***c*, **92***d*, a plurality of second walls **94***a*, **94***b*, **94***c*, **94***d*, and a connecting member **95**. The plurality of first walls **92***a*, **92***b*, **92***c*, **92***d* cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is rectangular. In some forms, each node **20***c* includes one or more first walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each first wall **92***a*, **92***b*, **92***c*, **92***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**3** such that the interior sides of the first walls **92***a*, **92***b*, **92***c*, **92***d* cooperate with each other to define a cavity **100** and end apertures **102** open to the cavity **100** through axial ends of the closed profile. In this way, the end apertures **102** may receive the respective frame member **22***a*, **22***b* of the frame **18**.

[0045] The plurality of second walls **94***a*, **94***b*, **94***c*, **94***d* cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the closed profile formed by the second walls **94***a*, **94***b*, **94***c*, **94***d* extend in a direction perpendicular from the closed profile formed by the first walls **92***a*, **92***b*, **92***c*, **92***d*. In some forms, each node **20***c* includes one or more second walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each second wall **94***a*, **94***b*, **94***c*, **94***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**4** such that the interior sides **106** of the second walls **94***a*, **94***b*, **94***c*, **92***d* cooperate with each other to define a cavity **110** and end apertures **112** open to the cavity **110** through respective axial ends of the node **20***c*. In this way, the end apertures **112** may receive the cross member **24***c* of the frame **18**. The connecting member **95** extends in a vertical direction and connects the first wall **94***a* and the second wall **92***b* to each other.

[0046] Each node **20***c* further includes a plurality of first ports **116***a*, a plurality of second ports **116***b*, first recessed channels (not shown), second recessed channels (not shown), at least one first locating feature **117***a*, at least one second locating feature **117***b*, an internal port **118** and a third port **120**. A bonding agent (e.g., adhesive such as epoxy) may be injected into the first ports **116***a* to bond the node **20***c* to the fame members **22***a*, **22***b*. A bonding agent may be injected into the second ports **116***b* to bond the node **20***c* to the cross member **24***c*. A bonding agent may be injected into the third port **120** and flow through the internal port **118** to further bond the node **20***c* to the frame **18** (i.e., bond the node **20***c* to both the cross member **24***c* and the respective frame member **22***a*, **22***b*). In the example illustrated, each first wall **92***a*, **92***c*, **92***d* includes at least one first port **116***a* and each second wall **94***b*, **94***c*, **94***d* includes at least one second port **116***b*. The structure and function of the first and second ports **116***a*, **116***b* may be similar or identical to the ports **46** described above, and therefore, will not be described again in detail.

[0047] The first recessed channels are formed in an interior side of each first wall **92***a*, **92***b*, **92***c*, **92***d* and intersect an outlet of the first port **116***a*. Similarly, the second recessed channels (not shown) are formed in an interior side (not shown) of each second wall **94***a*, **94***b*, **94***c*, **94***d* and intersect an outlet (not shown) of the second port **116***b*. The structure and function of the first and second recessed channels may be similar or identical to that of the recessed channels **48** described above, and therefore, will not be described again in detail.

[0048] In the example illustrated, one first locating feature **117***a* is formed in the first wall **92***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one first locating feature **117***b* is formed in the second wall **94***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. The structure and functions of the first and second locating features **117***a*, **117***b* may be similar or identical to the locating features **50** described above, and therefore, will not be described again in detail.

[0049] The internal port **118** extends from the first wall **92***b* through the connecting member **95**, to the second wall **94***a*. The third port **120** is located within the connecting member **95** and extends in a direction perpendicular to the internal port **118**. The third port **120** includes an outlet that opens through to the internal port **118** and an inlet that opens through to an exterior side **121** of the connecting member **95**. In this way, a bonding agent may be injected into the third port **120** to bond the node **20***c* to both the cross member **24***c* and the respective frame member **22***a*, **22***b*. The internal port **118** includes a size (e.g., a diameter) that is greater than a size (e.g., a diameter) of the third port **120**. Connecting members **126** may connect one or more of the first walls **92***a*, **92***b*, **92***c*, **92***d* to one or more of the second walls **94***a*, **94***b*, **94***c*, **94***d*.

[0050] With reference to FIGS. **4** and **5**D, each node **20***d* includes a plurality of walls **134***a* **134***b*, **134***c*, **134***d* that cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is square. In some forms, the node **20***d* includes one or more walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape.

Each wall **134**_a_, **134**_b_, **134**_c_, **134**_d_ has an interior side and an exterior side. The closed profile is disposed about an axis A**6** such that the interior sides of the walls **134**_a_, **134**_b_, **134**_c_, **134**_d_ cooperate with each other to define a cavity **140** and end apertures **142** open to the cavity **140** through respective axial ends of the node **20**_d_. In this way, the end apertures **142** may receive the cross member **24**_c_.

[0051] Each node **20**_d_ further includes a plurality of ports **146**, recessed channels, a plurality of locating features **150**, and at least one mounting structure **152**. A bonding agent (e.g., adhesive such as epoxy) may be injected into the ports **146** to bond the node **20**_d_ to the frame **18**. In the example illustrated, each wall **134**_a_, **134**_b_, **134**_c_, **134**_d_ includes at least one port **146**. The structure and function of the ports **146** may be similar or identical to the ports **46** described above, and therefore, will not be described again in detail.

[0052] The recessed channels are formed in an interior side of each wall **134**_a_, **134**_b_, **134**_c_, **134**_d_ and intersect an outlet (not shown) of the port **146**. The structure and function of the recessed channels may be similar or identical to that of the recessed channels **48** described above, and therefore, will not be described again in detail.

[0053] In the example illustrated, one first locating feature **150** is formed in the wall **134**_a_ and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one locating feature **150** is formed in the wall **134**_b_ and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. The structure and function of the locating features **150** may be similar or identical to the locating features **50** described above, and therefore, will not be described again in detail.

[0054] Each mounting structure **152** extends from one wall **134**_a_, **134**_b_, **134**_c_, **134**_d_ of the node **20**_d_ to a location where the workpiece **14** is mounted to an end face **156** of the mounting structure **152**. In the example illustrated, the mounting structure **152** includes a lattice array **157** having a plurality of beams or struts **158** oriented in a predetermined configuration. In the example illustrated, the beams **158** are solid and have a cylindrical shape. In some forms, the beams **158** are hollow and have a different shape (e.g., rectangular or square shape). In the example illustrated, the end face **156** is planar and is configured to be secured to a portion (e.g., a rail or cross member) of the workpiece **12**. In some forms, the end face **156** may be arcuate or any other suitable shape that is able to secure a portion of the workpiece **12** thereto.

[0055] With reference to FIGS. **4**, **5**E, and **7**, the nodes **20**_e_ attach a respective frame member **22**_a_, **22**_b_ to the node **20**_f_ via a respective rail **200** (FIG. **7**), and attach the frame members **22**_a_, **22**_b_ to each other via a respective cross member **24**_a_ (FIG. **4**). Each node **20**_e_ includes a plurality of first walls **202**_a_ **202**_b_, **202**_c_, **202**_d_, a plurality of second walls **204**_a_, **204**_b_, **204**_c_, **204**_d_, and a plurality of third walls, **206**_a_, **206**_b_, **206**_c_, **206**_d_. The plurality of first walls **202**_a_, **202**_b_, **202**_c_, **202**_d_ cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is rectangular. In some forms, each node **20**_e_ includes one or more first walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each first wall **202**_a_, **202**_b_, **202**_c_, **202**_d_ has an interior side and an exterior side. The closed profile is disposed about an axis A**10** such that the interior sides of the walls **202**_a_, **202**_b_, **202**_c_, **202**_d_ cooperate with each other to define a cavity **214** and an end aperture **216** open to the cavity **214** through an axial end of the node **20**_e_. In this way, the end aperture **216** may receive the cross member **24**_a_ of the frame **18**.

[0056] The plurality of second walls **204**_a_, **204**_b_, **204**_c_, **204**_d_ cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the closed profile formed by the second walls **204**_a_, **204**_b_, **204**_c_, **204**_d_ extend in a direction perpendicular from the closed profile formed by the first walls **202**_a_, **202**_b_, **202**_c_, **202**_d_. In some forms, each node **20**_e_ may include one or more second walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each second wall **204**_a_, **204**_b_, **204**_c_, **204**_d_ has an interior side and an exterior side. The closed profile is disposed about an axis A**11** such that the interior sides of the second walls

204*a*, **204***b*, **204***c*, **204***d* cooperate with each other to define a cavity **234** and end apertures **236** open to the cavity **234** through respective axial ends of the node **20***e*. In this way, the end apertures **236** may receive the respective frame member **22***a*, **22***b* of the frame **18**.

[0057] The plurality of third walls **206***a*, **206***b*, **206***c*, **206***d* cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the closed profile formed by the third walls **206***a*, **206***b*, **264***c*, **206***d* extends outward at an angle from the closed profile formed by the first walls **202***a*, **202***b*, **202***c*, **202***d* and at an angle from the closed profile formed by the second walls **204***a*, **204***b*, **204***c*, **204***d*. In some forms, each node **20***e* may include one or more third walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each third wall **206***a*, **206***b*, **206***c*, **206***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**12** such that the interior sides of the third walls **206***a*, **206***b*, **206***c*, **206***d* cooperate with each other to define a cavity **244** and an end aperture **246** open to the cavity **244** through an axial end of the node **20***e*. In this way, the end aperture **246** may receive the respective rail **200**. The axis A**12** extends at an angle relative to the axis A**11**. In the example illustrated, the angle is an acute angle.

[0058] Each node **20***e* further includes a plurality of first ports **256***a*, a plurality of second ports **256***b*, a plurality of third ports **256***c*, first recessed channel, second recessed channels, first locating features **258***a* and second locating features **258***b*. A bonding agent (e.g., adhesive such as epoxy) may be injected into the first ports **256***a* to bond the node **20***e* to the cross members **24***a*, a bonding agent may be injected into the second ports **256***b* to bond the node **20***e* to the frame members **22***a*, **22***b*, and a bonding agent may be injected into the third ports **256***c* to bond the node **20***e* to the rail **200**. In the example illustrated, each first wall **202***a*, **202***b*, **202***c*, **202***d* includes at least one first port **256***a*, each second wall **204***a*, **204***b*, **204***c*, **204***d* includes at least one second port **256***b*, and each third wall **206***a*, **206***b*, **206***c*, **206***d* includes at least one third port **256***c*. The structure and function of the first, second, and third ports **256***a*, **256***b*, **256***c* may be similar or identical to the ports **46** described above, and therefore, will not be described again in detail.

[0059] The first recessed channels are formed in an interior side of each first wall **202***a*, **202***b*, **202***c*, **202***d* and intersect an outlet of the first port **256***a*. The second recessed channels are formed in an interior side of each second wall **204***a*, **204***b*, **204***c*, **204***d* and intersect an outlet (not shown) of the second port **256***b*. The structure and function of the first and second recessed channels may be similar or identical to that of the recessed channels **48** described above, and therefore, will not be described again in detail.

[0060] In the example illustrated, one first locating feature **258***a* is formed in the first wall **202***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one first locating feature **258***a* is formed in the first wall **202***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. Similarly, one second locating feature **258***b* is formed in the second wall **204***a* and is vertically aligned with a corresponding aperture (not shown) formed in the frame **18**, and one second locating feature **258***b* is formed in the second wall **204***b* and is vertically aligned with another corresponding aperture (not shown) formed in the frame **18**. The structure and function of the first and second locating features **258***a*, **258***b* may be similar or identical to the locating features **50** described above, and therefore, will not be described again in detail. Connecting members **259** may connect one or more of the first walls **202***a*, **202***b*, **202***c*, **202***d* to one or more of the second walls **204***a*, **204***b*, **204***c*, **6204***d* and/or of the third walls **206***a*, **206***b*, **206***c*, **206***d*. In the example illustrated, mounting features **211** are formed in the second wall **204***d* of the nodes **204**. The mounting features **211** may be similar or identical to the mounting features **52** described above, and therefore, will not be described again in detail.

[0061] With reference to FIGS. **4**, **5**F, and **5**G, the node **20***f* is disposed between the frame members **22***a*, **22***b* and attaches the robot arm **14** to the nodes **20***e* via a respective rail **200**. The node **20***f* includes a hub **300**, a plurality of first walls **302***b*, **302***c*, **302***d*, a plurality of second walls **304***b*, **304***c*, **304***d*, a plurality of third walls **306***b*, **306***c*, **306***d*, and a plurality of fourth walls **308***b*, **308***c*,

308*d*. The hub **300** includes a wall **305** having a planar, upper surface **300***a* and a lower surface **300***b*. The upper surface **300***a* is raised from the frame **18** such that a predetermined distance is between the upper surface **300***a* and the frame **18**. The planar surface **300***a* includes a plurality of apertures **301** formed therethrough so that a second end **14***b* of the robot arm **14** is secured to the node **20***f*. The second end **14***b* of the robot arm **14** is secured to the planar surface **300***a* via an adapter (not shown). In some forms, the second end **14***b* of the robot arm **14** is secured directly to the planar surface **300***a* of the node **20***f* without an adapter.

[0062] With reference to FIGS. **5**F and **5**G, the plurality of first walls **302***b*, **302***c*, **302***d* extend from a respective end of the hub **300** and cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is square. In some forms, the node **20***f* includes one or more first walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each first wall **302***b*, **302***c*, **302***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**15** such that the interior sides of the walls **302***b*, **302***c*, **302***d* cooperate with each other to define a cavity **314** and an end aperture **316** open to the cavity **314**. In this way, the end aperture **316** may receive a respective rail **200**. The axis A**15** extends at an angle relative to a plane extending along the planar surface **300**. In the example illustrated, the angle is an acute angle.

[0063] The plurality of second walls **304***b*, **304***c*, **304***d* extend from a respective end of the hub **300** and cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is square. In some forms, the node **20***f* includes one or more second walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each second wall **304***b*, **304***c*, **304***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**17** such that the interior sides of the walls **304***b*, **304***c*, **304***d* cooperate with each other to define a cavity **324** and an end aperture **326** open to the cavity **324**. In this way, the end aperture **326** may receive a respective rail **200**. The axis A**17** extends at an angle relative to the plane extending along the planar surface **300**. In the example illustrated, the angle is an acute angle.

[0064] The plurality of third walls **306***b*, **306***c*, **306***d* extend from the hub **300** and cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is square. In some forms, the node **20***f* includes one or more third walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each third wall **306***b*, **306***c*, **306***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**18** such that the interior sides of the walls **306***b*, **306***c*, **306***d* cooperate with each other to define a cavity **334** and an end aperture **336** open to the cavity **334**. In this way, the end aperture **336** may receive a respective rail **200**. The axis A**18** extends at an angle relative to the plane extending along the planar surface **300**. In the example illustrated, the angle is an acute angle.

[0065] The plurality of fourth walls **308***b*, **308***c*, **308***d* extend from a respective end of the hub **300** and cooperate with each other to define a closed profile having an annular shape. In the example illustrated, the annular shape is square. In some forms, the node **20***f* includes one or more fourth walls such that the annular shape is circular, pentagonal, hexagonal, or any other suitable shape. Each fourth wall **308***b*, **308***c*, **308***d* has an interior side and an exterior side. The closed profile is disposed about an axis A**19** such that the interior sides of the walls **308***b*, **3080**, **308***d* cooperate with each other to define a cavity **344** and an end aperture **346** open to the cavity **344**. In this way, the end aperture **346** may receive a respective rail **200**. The axis A**19** extends at an angle relative to the plane extending along the planar surface **300**. In the example illustrated, the angle is an acute angle.

[0066] The node **20***f* further includes a plurality of first ports **356***a*, a plurality of second ports **356***b*, a plurality of third ports **356***c*, and a plurality of fourth ports **356***d*. A bonding agent (e.g., adhesive such as epoxy) may be injected into the first ports **356***a* to bond the node **20***f* to the respective rail **200**, a bonding agent may be injected into the second ports **356***b* to bond the node

20*f* to the respective rail **200**, a bonding agent may be injected into the third ports **356***c* to bond the node **20***f* to the respective rail **200**, and a bonding agent may be injected into the fourth ports **356***d* to bond the node **20***f* to the respective rail **200**. In the example illustrated, each first wall **302***b*, **302***c*, **302***d* includes at least one first port **356***a*, each second wall **304***b*, **304***c*, **304***d* includes at least one second port **356***b*, each third wall **306***b*, **306***c*, **306***d* includes at least one third port **356***c*, and each fourth wall **308***b*, **308***c*, **308***d* includes at least one fourth port **356***d*. The structure and function of the first, second, third, and fourth ports **356***a*, **356***b*, **356***c*, **356***d* may be similar or identical to the ports **46** described above, and therefore, will not be described again in detail.

[0067] One first port **357***a* includes an inlet that opens through to the upper surface **300***a* and an outlet that opens through to the cavity **314**. One second port **357***b* includes an inlet that opens through to the upper surface **300***a* and an outlet that opens through to the cavity **324**. One third port **357***c* includes an inlet that opens through to the upper surface **300***a* and an outlet that opens through to the cavity **334**. One fourth port **357***d* includes an inlet that opens through to the planar surface **300** and an outlet that opens through to the cavity **344**.

[0068] With reference to FIG. **11**, a method for assembling the robotic end effector **16** will be described in detail. First, at **404**, the nodes **20***a*, **20***b*, **20***c*, **20***e*, **51***b* are connected to a respective frame member **22***a*, **22***b*. That is, in the example illustrated, the nodes **20***a*, **20***b*, **20***c*, **20***e*, **51***b* are slid onto the respective frame members **22***a*, **22***b* in the following order: the nodes **20***e* are slid onto the frame members **22***a*, **22***b*, then two of the nodes **20***a* are slid onto respective ends **23***b* of the frame members **22***a*, **22***b*, then the nodes **51***b* are slid onto the frame members **22***a*, **22***b*, then two more of the nodes **20***a* are slid onto the frame members **22***a*, **22***b*, then the nodes **20***c* are slid onto the frame members **22***a*, **22***b*, then the nodes **20***b* are slid onto the frame members **22***a*, **22***b*. It should be understood that the nodes **20***a*, **20***b*, **20***c*, **20***e*, **51***b* may take a different form or pattern than what's shown in the figures. That is, nodes **20***c* may be located closer to the nodes **20***e* than nodes **51***b*, for example. In another form, two of the nodes **20***a* may be removed from the frame members **22***a*, **22***b*.

[0069] Then, at **408**, the cross members **24***a*, **24***b*, **24***c* are connected to respective nodes **20***b*, **20***c*, **20***e*. That is, the cross members **24***a* are connected to the nodes **20***e*, the cross member **24***b* is connected to the nodes **20***b* and the cross member **24***c* is connected to the nodes **20***c*. Then, the nodes **20***d* are slid onto the cross member **24***c* and two more of the nodes **20***a* are slid onto the cross member **24***b*.

[0070] Then, at **412**, the nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **51***a*, **51***b* are held in place by respective pins while a bonding agent is injected into ports of the nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f*, **51***a*, **51***b* to bond the nodes **20***a*, **20***b*, **20***c*, **20***d*, **20***e*, **20***f*, **51***a*, **51***b* to the frame **18**. The robotic end effector **16** of the present disclosure provides the benefit of securing the frame **18** together using various nodes and a bonding agent.

[0071] Unless otherwise expressly indicated herein, all numerical values indicating mechanical/thermal properties, compositional percentages, dimensions and/or tolerances, or other characteristics are to be understood as modified by the word “about” or “approximately” in describing the scope of the present disclosure. This modification is desired for various reasons including industrial practice, material, manufacturing, and assembly tolerances, and testing capability.

[0072] As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0073] In this application, the term “controller” and/or “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores

code executed by the processor circuit; other suitable hardware components (e.g., op amp circuit integrator as part of the heat flux data module) that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.
[0074] The term memory is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).
[0075] The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general-purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.
[0076] The description of the disclosure is merely exemplary in nature and, thus, variations that do not depart from the substance of the disclosure are intended to be within the scope of the disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure.

Claims

1. A robotic end effector comprising: a first node including at least one wall, the at least one wall having a corresponding interior side and a corresponding exterior side, wherein the at least one wall defines a closed profile disposed about an axis such that the corresponding interior side of the at least one wall defines a cavity and the at least one wall defines an end aperture open to the cavity through an axial end of the first node, the end aperture being configured to receive a first frame member, wherein the at least one wall includes a first port having a first outlet open through the corresponding interior side of the at least one wall and a first inlet open through the corresponding exterior side of the at least one wall, wherein corresponding the interior side of the at least one wall defines a plurality of first recessed channels that intersect the first outlet of the first port.
2. The robotic end effector of claim 1, wherein the first node is made of plastic.
3. The robotic end effector of claim 1, wherein the corresponding interior side of the first node defines an interior surface, and wherein the first recessed channels are formed in the interior surface.
4. The robotic end effector of claim 1, wherein the first node includes a locating feature that is adjacent the first port and configured to facilitate positioning of the first node relative to the first frame member, and wherein the locating feature includes a second outlet open through the corresponding interior side of the first node and a second inlet end open through to the corresponding exterior side of the first node.
5. The robotic end effector of claim 4, wherein the first recessed channels are spaced apart from the second outlet of the locating feature and extend around the second outlet of the locating feature.
6. The robotic end effector of claim 1, wherein the first node further includes: a second port that includes a second outlet open through the corresponding interior side of the first node and a second inlet open through the corresponding exterior side of the first node; and second recessed channels formed in the corresponding interior side of the first node and intersecting the second outlet of the second port.

7. The robotic end effector of claim 6, wherein the second recessed channels have a pattern that is different from a pattern of the first recessed channels.
8. The robotic end effector of claim 1, wherein the first node includes a mounting feature that is adjacent to the first port and configured to mount a structure to the first node.
9. The robotic end effector of claim 8, wherein the mounting feature comprises a plurality of threaded apertures.
10. The robotic end effector of claim 1, further comprising the first frame member, wherein a bonding agent flows through the first port to the first recessed channels to attach the first node to the first frame member.
11. The robotic end effector of claim 10, further comprising a second node that is spaced apart from the first node and attached to the first frame member.
12. A robotic end effector comprising: a frame including a first member and a second member that extends in a direction that is different than the first member; and a first node including a first interior side, a second interior side, and an exterior side, the first interior side faces the first member and defines a first interior cavity that receives the first member, the second interior side faces the second member and defines a second interior cavity that receives the second member, the exterior side faces away from the frame, the first node further includes: a first port and first recessed channels, the first port includes a first outlet that opens through to the first interior side of the first node and a first inlet that opens through to the exterior side of the first node, the first recessed channels formed in the first interior side of the first node proximate the first member and intersecting the first outlet of the first port; and a second port and second recessed channels, the second port includes a second outlet that opens through to the second interior side of the first node and a second inlet that opens through to the exterior side of the first node, the second recessed channels are formed in the second interior side of the first node proximate the second member and intersecting the second outlet of the second port.
13. The robotic end effector of claim 12, wherein the second recessed channels form a pattern that is different from a pattern of the first recessed channels.
14. The robotic end effector of claim 12, further comprising a second node spaced apart from the first node and attached to the frame.
15. The robotic end effector of claim 12, wherein the first node includes a mounting feature that is adjacent to the first port and configured to mount a structure to the first node.
16. The robotic end effector of claim 12, wherein the first node further includes: a third port that includes a third outlet and a third inlet, the third inlet opens through to the exterior side of the first node; and an internal port located within the first node and extending from the first interior side of the first node to the second interior side of the first node, the internal port in fluid communication with the third outlet of the third port.
17. The robotic end effector of claim 16, wherein the internal port includes a diameter that is greater than a diameter of the third port.
18. The robotic end effector of claim 12, wherein the first node includes a locating feature that is adjacent to one of the first and second ports and that facilitates positioning of the first node relative to the frame, and wherein the locating feature includes a third outlet that opens through to one of the first and second interior sides of the first node and a third inlet that opens through to the exterior side of the first node.
19. The robotic end effector of claim 18, wherein one of the first and second recessed channels are spaced apart from the third outlet of the locating feature and extend around the third outlet of the locating feature.
20. A method of assembling a robotic end effector, the method comprising: inserting a first frame member into a first interior cavity of a node such that first recessed channels defined by a first interior surface of the first interior cavity are proximate the first frame member; inserting a second frame member into a second interior cavity of the node such that second recessed channels defined

by a second interior surface of the second interior cavity are proximate the second frame member; injecting a first bonding agent into a first port of the node such that it flows along the first recessed channels in the first interior surface; and injecting a second bonding agent into a second port of the node such that it flows along the second recessed channels in the second interior surface.